A GEOMETRIC POINT OF AN ETALE NEIGHBOURHOOD LIFTS TO SOME MEMBER OF ANY ETALE COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The Xy-pic xy environment with its curved arrows is required because the counted proof displays a cartesian square with the section sigma drawn as a curved arrow. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The same-chapter referenced proposition is located at its \label line inside the pinned chapter file.

Allowed standard prerequisites (editorial).

- Etale neighbourhood of a geometric point, and etale covering of a scheme. Same chapter, the definition of an etale neighbourhood (U, ubar) with U etale over S, and the standard notion of an etale covering family. Supplies the objects and the surjectivity U = union of the images of the U_i; the construction of the lift is the counted content.
- Structure of etale morphisms over an algebraically closed field. Same chapter, Proposition labelled proposition-etale-morphisms (source line 2592, Stacks tag 03PC); invoked at proof lines 3329-3333. Supplies that a base change of an etale morphism is etale and that a scheme etale over the spectrum of an algebraically closed field is a disjoint union of trivial extensions; the section argument and the lifting conclusion are retained in the counted span.

Source reference. Original proposition 03PC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2592–2592. Same-chapter pointer invoked twice, at proof lines 3330 and 3333, for base change of etale morphisms and for the shape of a scheme etale over an algebraically closed field; the cartesian square, the nonemptyness argument and the section construction are retained verbatim in the counted span.

Lemma 0.1. *Let S be a scheme. Let $\overline{s}$ be a geometric point of S. Let $(U, \overline{u})$ be an étale neighborhood of $\overline{s}$. Let $\mathcal{U} = \{\varphi_i : U_i \to U\}_{i \in I}$ be an étale covering. Then there exist $i \in I$ and $\overline{u}_i : \overline{s} \to U_i$ such that $\varphi_i : (U_i, \overline{u}_i) \to (U, \overline{u})$ is a morphism of étale neighborhoods.*

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Proof. As $U = \bigcup_{i \in I} \varphi_i(U_i)$, the fibre product $\overline{s} \times_{\overline{u}, U, \varphi_i} U_i$ is not empty for some i. Then look at the cartesian diagram

$$\begin{array}{ccc} \overline{s} \times_{\overline{u}, U, \varphi_i} U_i & \xrightarrow[\mathrm{pr}_2]{} & U_i \\ {\scriptstyle \sigma}\big(\big\downarrow{\scriptstyle \mathrm{pr}_1} & & \big\downarrow{\scriptstyle \varphi_i} \\ \mathrm{Spec}(k) = \overline{s} & \xrightarrow{\overline{u}} & U \end{array}$$

The projection pr_1 is the base change of an étale morphisms so it is étale, see Proposition 03PC. Therefore, $\overline{s} \times_{\overline{u}, U, \varphi_i} U_i$ is a disjoint union of finite separable extensions of k, by Proposition 03PC. Here $\overline{s} = \mathrm{Spec}(k)$. But k is algebraically closed, so all these extensions are trivial, and there exists a section σ of pr_1. The composition $\mathrm{pr}_2 \circ \sigma$ gives a map compatible with $\overline{u}$. $\square$